PIPELINE AUDIT AND HEALTH REPORT

Electronics: SMT circuit-board line

Paste printer, chip shooter and fine-pitch placer feed a 4-minute reflow oven, then AOI inspection (a custom quality_gate node) sends bad boards to a rework bench or scrap. The chip shooter is the slowest and fails often, so it should show up as the constraint. Try the What-if tab on it.

Generated 2026-10-13 19:48 | simulated 8.0 h (warm-up 30 min) | seed 12 | 10 replications

HEALTH SCORE

69 /100

Grade D

At risk: significant constraints or fragility; fix the high-severity findings first.

Critical 0 | High 1 | Medium 4 | Low 0 | Info 1

The score starts at 100 and loses points for each finding. The deductions are listed in Appendix B.

THROUGHPUT	COMPLETED	ARRIVALS LOST
71.9 units/h	576 units	0.0%

AVERAGE WIP	CYCLE TIME	BOTTLENECK
13.5 units	674 s	pnp_chip

Executive summary

Primary bottleneck: 'pnp_chip' (77% busy, 19% down, index 103). Sources are held back up to 31% of the time. Run what-if to measure which lever recovers the most throughput.

What to fix first

- High (pnp_chip): Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: 10% faster processing at 'pnp_chip' (+6.6% throughput).
- Medium (supply): The line cannot absorb the offered supply, so everything upstream of the constraint spends time waiting.
- Medium (printer): Speed up or add capacity downstream of this station, or enlarge the buffer after it.
- Medium (pnp_chip): Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.

Best measured improvement

Best lever: 10% faster processing at 'pnp_chip' -> +4.9 units/hour (6.6%), measured over 3 seeds with common random numbers.

Findings

Each finding names where the problem is, why it matters, and what to do. Severity reflects the impact on throughput, correctness or confidence in the results.

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
High	Capacity	**pnp_chip**	The bottleneck is 96% utilised (busy or down). There is no slack: any disturbance directly cuts output. Action: Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: 10% faster processing at 'pnp_chip' (+6.6% throughput).	-8
Medium	Flow	**supply**	Upstream supply is held back up to 31% of the time by a full line. Action: The line cannot absorb the offered supply, so everything upstream of the constraint spends time waiting.	-5
Medium	Flow	**printer**	Blocked 40% of the time: finished parts cannot leave because the next buffer is full. Action: Speed up or add capacity downstream of this station, or enlarge the buffer after it.	-3
Medium	Reliability	**pnp_chip**	Down 19% of the time (15 failures). Action: Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.	-3
Medium	Reliability	**printer, pnp_chip, pnp_fine**	Every unit passes through these stations, and they can fail. A single failure stops the whole plant. Action: Add a parallel machine or a buffer of finished work in front of the downstream stations so they can ride through a repair.	-12
Info	Capacity	**pnp_chip**	Constraint of the line: 96% utilised. Action: Improvements elsewhere will not raise throughput until this station is relieved. Measured best lever: 10% faster processing at 'pnp_chip' (+6.6% throughput).	

Performance by node

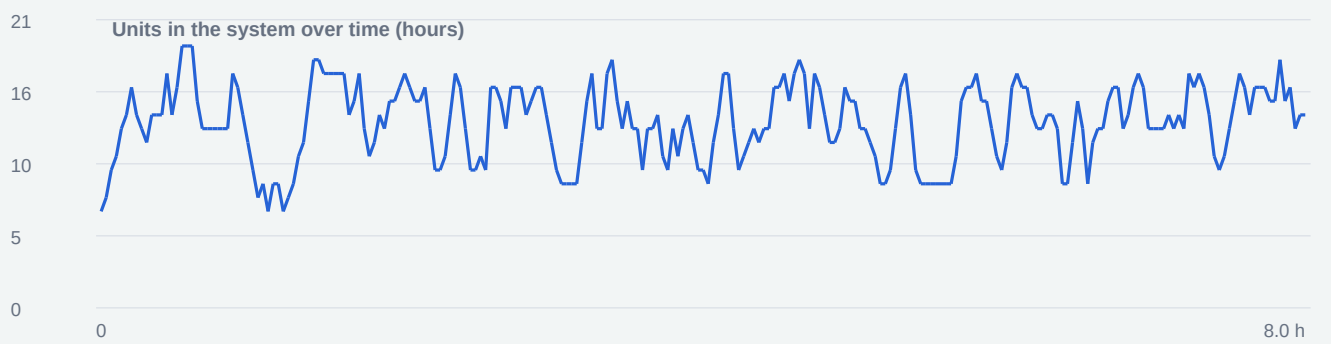
Share of the measured window each node spent busy, blocked (finished part cannot leave), down (failed) and starved (waiting for work).

Legend: green = busy, amber = blocked, red = down, grey = starved or idle.

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
boards	source	-	-	-	-	-
printer	machine		50.2	40.5	0.4	2.8 / 4
pnp_chip bottleneck	machine		76.9	3.0	18.6	3.7 / 4
pnp_fine	machine		60.2	0.0	5.6	0.9 / 4
reflow	delay		87.8	0.0	0.0	4.8 / 8
aoi	quality_gate		25.8	0.0	0.0	0.3 / 2
rework_bench	machine		24.6	0.0	0.0	0.3 / 2

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
test	machine		40.0	0.0	0.0	0.4 / 2
shipped	sink	-	-	-	-	-
scrap	sink	-	-	-	-	-

Work in process over time



Reliability and robustness

The pipeline was simulated 10 times with different random seeds. A wide interval means the result depends strongly on luck.

METRIC	MEAN	95% CONFIDENCE	STD DEV	RELATIVE
Throughput	76.96 units/h	+- 3.69	5.16	4.8%
Average WIP	13.28 units	+- 0.29	0.40	2.2%
Cycle time	619.95 s	+- 34.84	48.71	5.6%
Arrivals lost	0.00 %	+- 0.00	0.00	0.0%

No replication deadlocked.

Failure model

NODE	MTBF	MTTR	AVAILABILITY	OBSERVED DOWN	FAILURES
printer	90 min	4.0 min	95.7%	0.4%	1
pnp_chip	40 min	5.0 min	88.9%	18.6%	15
pnp_fine	120 min	7.0 min	94.5%	5.6%	5

Single points of failure

All units must pass through: printer, pnp_chip, pnp_fine, aoi. If one of these stops, output stops.

Model validity checks

CHECK	RESULT	DETAIL
Unit conservation	PASS	created 592 = completed 576 + scrapped 2 + lost 0 + in system 14
Little's Law	PASS	L = 13.50, lambda x W = 13.51 (gap 0.1%)
Run length	PASS	window = 40 x cycle time (30x or more recommended)
Warm-up	PASS	first 30 min excluded from statistics
Deadlock-free	PASS	reference run completed

What-if analysis

Each lever was applied to one of the most constrained stations and the whole pipeline re-simulated over at least 8 hours with 3 identical random seeds per scenario. Baseline throughput: 73.6 units/h.

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
pnp_chip	10% faster processing	78.5	**+4.9**	+6.6%
pnp_chip	20% shorter repairs	75.1	**+1.5**	+2.1%
pnp_chip	+2 buffer slots	74.0	**+0.4**	+0.6%
pnp_fine	20% shorter repairs	74.0	**+0.4**	+0.5%
printer	+2 buffer slots	74.0	**+0.4**	+0.5%
pnp_fine	+2 buffer slots	73.9	**+0.3**	+0.4%
pnp_fine	10% faster processing	73.8	**+0.2**	+0.3%
printer	10% faster processing	73.8	**+0.2**	+0.2%
printer	20% shorter repairs	73.7	**+0.1**	+0.1%

Best lever: 10% faster processing at 'pnp_chip' -> +4.9 units/hour (6.6%), measured over 3 seeds with common random numbers.

Appendix A: node inventory

Effective parameters after node-type templates are applied. Parameters not listed use the node type's defaults.

NODE	TYPE	EFFECTIVE PARAMETERS
boards	source	interarrival=exponential(36), on_blocked=wait
printer	machine	capacity=3, mtbf=5400, mttr=240, process=normal(25,3)
pnp_chip	machine	capacity=3, mtbf=2400, mttr=300, on_failure=restart, process=normal(38,4)
pnp_fine	machine	capacity=3, mtbf=7200, mttr=420, process=normal(30,3)
reflow	delay	capacity=8, transit=240

NODE	TYPE	EFFECTIVE PARAMETERS
aoi	quality_gate	capacity=4, max_reworks=1, pass_to=test, process=normal(12,2), rework_to=rework_bench, scrap_to=scrap, threshold=0.06
rework_bench	machine	capacity=3, process=normal(180,40)
test	machine	capacity=3, process=normal(20,3)
shipped	sink	(defaults)
scrap	sink	counts_as=scrapped

Appendix B: how the score is computed

Health score = 100 minus the sum of the penalties below, never less than 0. Grades: A 90 and above, B 80 to 89, C 70 to 79, D 60 to 69, F below 60. Penalties are capped per category so that one repeated issue cannot dominate the score. Informational findings carry no penalty.

SEVERITY	WHERE	FINDING	POINTS
High	**pnp_chip**	The bottleneck is 96% utilised (busy or down). There is no slack: any disturbance directly cuts output.	-8
Medium	**supply**	Upstream supply is held back up to 31% of the time by a full line.	-5
Medium	**printer**	Blocked 40% of the time: finished parts cannot leave because the next buffer is full.	-3
Medium	**pnp_chip**	Down 19% of the time (15 failures).	-3
Medium	**printer, pnp_chip, pnp_fine**	Every unit passes through these stations, and they can fail. A single failure stops the whole plant.	-12
	Total deductions		**-31**
	Health score		**69**

Method: the pipeline is simulated as a discrete-event system; the bottleneck index is utilisation (busy plus down time) with small weights for starved downstream and blocked upstream neighbours. What-if results are measured, not estimated. Confidence intervals use Student's t over independent replications.